

Software Requirements Specification

Student Startup Crowdfunding Platform — PS#40

Software Engineering Laboratory

Version 1.0

Document Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements, actors, use-case behavior, system scope, architecture expectations, and traceability information for the Student Startup Crowdfunding Platform. The document is based on the project requirements and UC-05 release-tranche flow prepared for PS#40.

1. Introduction

1.1 Purpose

The platform supports student founders who raise funds through crowdfunding campaigns and campaign backers who pledge money in return for defined reward tiers. Faculty Committee review controls milestone-based release of funds.

1.2 Scope

The system covers campaign creation and eligibility validation, campaign browsing, reward-tier selection and pledging, milestone evidence review, tranche-based fund release, and automatic cancellation/refunding when a campaign fails to reach its goal by its deadline. Payment processing is delegated to a tokenized payment gateway.

1.3 Stakeholders and Actors

- Student Founder — creates and manages a campaign and submits milestone evidence.
- Campaign Backer — browses active campaigns, selects reward tiers, and submits pledges.
- Faculty Committee — reviews milestone evidence and records approval, rejection, or requests for more information.
- Payment Gateway — securely processes pledge and disbursement transactions.

2. System Overview

The platform coordinates crowdfunding activities between student founders, backers, faculty reviewers, and a payment gateway. Funds are held until the relevant milestone has been reviewed and approved. The system records review decisions, payment results, and notifications so that the funding process is traceable.

3. Functional Requirements

ID	Requirement	Priority
FR-001	Collect backer pledges and disburse funds to the student venture in tranches only after verified milestones.	High
FR-002	Campaign Backer browses active campaigns, selects a reward tier, and submits a pledge meeting the tier minimum.	High
FR-003	Faculty Committee reviews milestone evidence and records Approve/Reject/Request-More-Info with comments.	High
FR-004	Student Founder creates a campaign with goal, deadline, milestone plan, reward tiers, and eligibility validation.	High
FR-005	Automatically cancel the campaign and refund pledges if the funding goal is not met by the deadline.	Medium

4. Non-Functional Requirements

ID	Requirement	Priority
NFR-001	Payment gateway integration shall use PCI-DSS-compliant/tokenized processing; the platform shall not store or transmit raw payment data. P95 pledge latency shall be below 2 seconds.	High

ID	Requirement	Priority
NFR-002	The platform shall provide at least 99.9% uptime during active pledge windows and preserve durable records through a single-node failure.	High

5. Use Cases

5.1 Primary Use Cases

UC-01 Create Campaign — Student Founder creates a campaign and provides the funding goal, deadline, milestone plan, reward tiers, and eligibility information.

UC-02 Pledge to Campaign — Campaign Backer browses active campaigns, selects a reward tier, and submits a qualifying pledge.

UC-03 Review Milestone Evidence — Faculty Committee reviews evidence and records an approval, rejection, or request for more information.

UC-04 Cancel Campaign and Refund Pledges — the system cancels a campaign and initiates refunds when the deadline passes without the required goal being met.

UC-05 Release Tranche Funds — after an authorized milestone approval, the system calculates the tranche, requests disbursement through the Payment Gateway, records the result, and notifies relevant users.

5.2 UC-05 Release Tranche Funds

Primary actor: Faculty Committee

Secondary actors: Student Founder, Payment Gateway

Related requirements: FR-001, FR-003

Preconditions: A milestone has reached its required threshold; milestone evidence has been submitted; the reviewer is authorized.

Main success flow: (1) Founder submits milestone evidence. (2) System records the submission. (3) Committee reviews the evidence. (4) Committee approves the milestone. (5) System records the decision with reviewer and timestamp. (6) System marks the milestone approved and calculates the tranche. (7) System sends a disbursement request to the Payment Gateway. (8) Gateway confirms the payment. (9) System records the payment and marks the tranche released. (10) System notifies the founder and relevant backers.

Alternate flow: If the committee rejects the evidence or requests more information, no funds are transferred. The system records the decision/comments and retains the funds until a valid approval is obtained.

Business rules: Only one final authorized decision determines the outcome; earlier decisions remain in the audit history. Rejected funds remain held. An approved tranche is released within the specified business-time window.

6. System Behavior and Business Rules

- Pledges must satisfy the minimum amount associated with the selected reward tier.
- Milestone funds cannot be released without an authorized approval.
- Review decisions must be auditable with reviewer and timestamp information.
- If a milestone is rejected or more information is requested, funds remain held.
- If a campaign fails to reach its goal by the deadline, the campaign is cancelled and pledges are refunded.
- Raw payment credentials are not stored or transmitted by the platform.

7. System Architecture Overview

The proposed architecture is a layered web application. The presentation layer provides interfaces for founders, backers, and faculty reviewers. The application/service layer implements campaign, pledge, milestone, review, refund, notification, and disbursement logic. A persistent data layer stores campaign, pledge, milestone, review, and audit records. Payment operations are delegated to an external tokenized Payment Gateway. The architecture should support durable records and recovery from a single-node failure.

8. UML and Design Artifacts

The project repository contains the Use Case Diagram and Activity Diagram for the system. The Use Case Diagram models the principal actors and required interactions, including an «include» relationship for recording a review decision and an «extend» relationship for requesting more information during milestone review. The Activity Diagram models the UC-05 tranche-release flow and its alternate outcomes.

9. Exception and Alternate Flow

For UC-05, a rejected milestone or a request for additional information stops the disbursement path. No funds are released unless the milestone receives the required authorized approval. Formal rejection and appeal handling are outside the core release flow and may be handled by a separate process.

10. Traceability

The Requirements Traceability Matrix (RTM.xlsx) in the repository maps FR-001 through FR-005 and NFR-001 through NFR-002 to their related use cases/design flows, acceptance criteria, and planned test cases. The RTM is maintained as the central traceability artifact.

11. Acceptance Summary

The system is considered functionally aligned with this SRS when all five functional requirements are demonstrably implemented/tested, both non-functional requirements have corresponding design/verification evidence, milestone approval controls fund release, unsuccessful campaigns trigger refunds, and the RTM provides traceability from requirements to tests.

12. Conclusion

This SRS establishes the baseline requirements and behavior for the Student Startup Crowdfunding Platform. Subsequent architecture, implementation, and testing activities should trace back to the requirements and acceptance criteria defined here.